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Roll No. 3112413058

324451(25)

B. E. (Fourth Semester)
EXAMINATION, April-May, 2015
(New Course)
(Branch : Electrical Engineering)
ANALOG ELECTRONICS

Time : Three Hours] [*Maximum Marks : 80*

[*Minimum Pass Marks : 28*

Note : Part (a) of each question carries 2 marks and is compulsory. Attempt any *two* parts from (b), (c) and (d) which carry 7 marks each.

Unit—I

1. (a) Define the following terms operating point. 2

(b) Explain in detail different types of stabilization techniques and compensation techniques. 7

(c) Describe base with emitter feedback in detail. Why is this method failure achieve stability ? Calculate stability factor for it. 7

(d) In the circuit shown $V_{CC} = 24$ V, $R_C = 10$ K and $R_E = 270\ \Omega$. If a silicon transistor is used

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with B = 45 and if under quiescent condition $V_{CE} = 5$V , determine. 7

(i) R_B and

(ii) The stability factor 5.

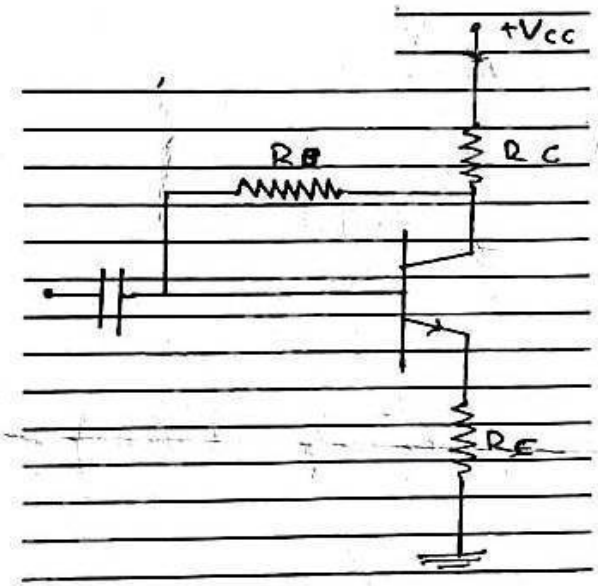

Fig. 1

Unit—II

2. (a) What is meant by Cascading ? 2

(b) Draw the low frequency equivalent circuit of an amplifier using *h*-parameters and derive expression for current gain A_i, input resistance and voltage gain. 7

(c) Compare CE, CC and CB configuration of transistor. 7

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(d) A CE amplifier with bypassed R_E using a PNP transistor is shown in Fig 2. Calculate A_I, A_{IS}, R_i, R_i', A_v, A_{vs}, R_o and R_o' using approximate analysis. Assume $h_{ie} = 1.1\ k\Omega$ and $h_{fe} = 50$. 7

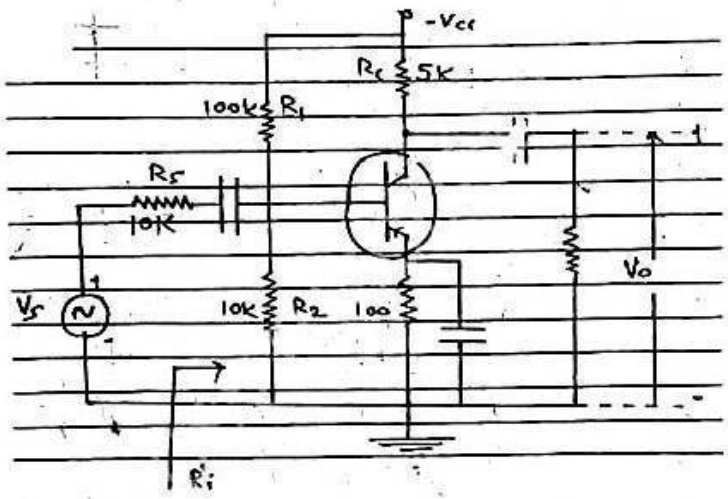

Fig. 2

Unit—III

3. (a) Define transconductance g_m. 2

(b) Find the relationship between low frequency and high frequency parameters. 7

(c) Draw the circuit and derive the expression for CE short circuit current gain A_i in terms of at any frequency f and f_B of the BJT. 7

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(d) At $I_C = 1$ mA and $V_{CE} = 10$ V, a certain transistor data shows $C_C = C_{b'c} = 3$pF, $h_{fe} = 200$ and $\omega_T = -500$ M rad/ sec. Calculate g_m, $r_{b'e}$, $C_e = C_{b'e}$ and ω_β. 7

Unit—IV

4. (a) Define the following : 2

(i) Non-linear distortion

(ii) Frequency distortion

(iii) Phase shift distortion

(iv) Bandwidth of amplifier

(b) Define noise and explain the noise figure. Derive the necessary equation for noise figure. 7

(c) Compare direct coupling, RC coupling and transformer coupling. 7

(d) An amplifier consists of 3 identical stages in cascade, the bandwidth of overall amplifier extend from 20 Hz to 20 kHz. Calculate the bandwidth of individual stage. 7

Unit—V

5. (a) Draw the block diagram of feedback amplifier which gives low output resistance and low input resistance. 2

(b) Explain the concept of feedback in amplifier. List the important properties of negative feedback. 7

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(c) Draw the circuit diagram of voltage shunt feedback amplifier and derive an expression for input and output impedance. 7

(d) The transistor shown in Fig. below has the following h-parameters; h_{ie} = 1.1 kΩ, h_{fe} = 50, $h_{re} = h_{oe} = 0$. Calculate A_{vf}, R_{if} and R'_{of}. 7

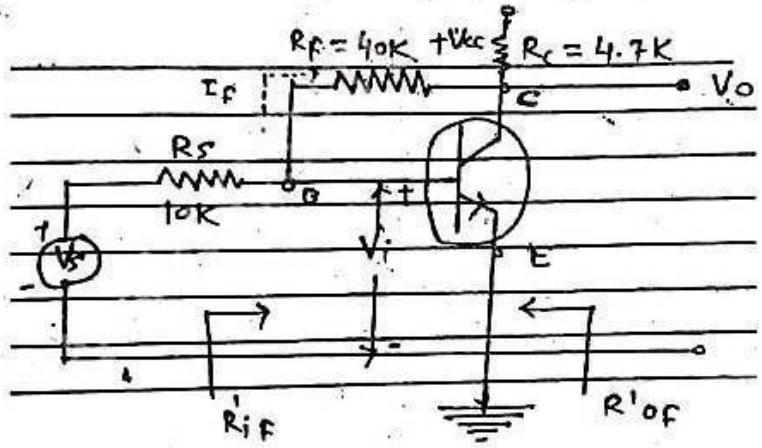

Fig. 3

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